TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM FOUR INTER ISLAMIC MOCK EXAMINATION

CHEMISTRY 2A

(ACTUAL PRACTICAL)

32/2A

(For Both School and Private Candidates)

TIME: 2:30 HOURS **Tuesday, 27th August 2024 a.m.**

wazaelimu.com

Instructions

1. This paper consists of two (2) questions. Answer **both** questions.
2. Each question carries twenty five (25) marks.
3. Qualitative analysis guide sheet authorized by NECTA and non-programmable calculators may be used.
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).
6. The following constants may be used:
   - Atomic masses: H=1, C=12, O=16, Na=23, Mg=24, Cl=35.5, Al=27, p = 31, Ca= 40.
   - Avogadro's number = 6.02×10^{23}
   - Standard Pressure = 760mmHg.
   - Standard Temperature = 273k
   - $1L=1dm^3 = 1000cm^3$
   - GMV = $22.4dm^3$
   - 1 Faraday = 96500C

1. You are provided with the following:

   Solution NN: Prepared when form one student accidentally poured pellets of Sodium hydroxide (NaOH) with unknown mass in a container containing distilled water. The resulting solution was found to have a volume of 2 litres.

   Solution MM: Containing 360g of pentahydrate Oxalic acid ($H_2C_2O_4.5H_2O$) dissolved to make 2 litres of the solution.

   Phenolphthalein indicator and methyl orange indicator.

   Procedures:

   (i) Measure exactly 75cm^3 of the acid by using measuring cylinder and then transfer it into an empty clean beaker.

   (ii) Measure exactly 75cm^3 of distilled water and then put them into the solution of the acid in part (i) above.

   (iii) Stir well the solution mixture in part (ii) above by means of clean dry glass rod.

   (iv) Put the mixture into the burette.

   (v) Pipette 20cm^2 or 25cm^3 of the solution NN into a titration flask. Add two drops or three (3) drops of the suitable indicator from the given ones. Titrate the base solution against the mixture from the burette until an end point is reached.

   (vi) Record your results in a tabular form.

   Questions:

   (a) Why is it important to clean a titration flask at the end of each titration before starting the subsequent titration?

   (b) State the role of the white tile or white paper on which the titration flask with its content is placed during titration.

   (c) What was the volume of the pipette used?

   (d) Calculate the average volume of the acid used.

   (e) Indicating all the state symbols, write a balanced chemical equation for the reaction occurred.

   (f) Write the net ionic equation for the reaction involved.

   (g) Showing your procedures clearly, calculate the exact mass of the base poured by the student.

   (h) Classify the acid based on its strength, basicity and origin.

   (i) Giving reason, defend the choice you made in using indicator that you have used in the titration above.

2. You are provided with the salt sample M containing one cation and one anion. Carryout the experiments described in the table below. Record carefully your observation, make appropriate inferences and finally identify the cation and anion present in the sample M.

	EXPERIMENT	OBSERVATION	INFERENCE
(a)	Observe the appearance of sample M		
(b)	Heat a little amount of sample M in a dry test tube		
(c)	Dissolve the little amount of sample M in water then divide the solution into three portions;		
	(i) To the first portion add NaOH solution till in excess.		
	(ii) To the second portion add dilute H_2SO_4 and then a solution of $FeSO_4$ followed by addition of Concentrated H_2SO_4.		
	(iii) To the third portion add a small amount of potassium hexacyanoferrate (II) solution		

Conclusion:

(i) The cation in sample M is _____

(ii) The anion in sample M is _____

(iii) The molecular formular for sample M is _____

(iv) Write a balanced chemical equation for the reaction occurred in experiment (b) above.

Wabillah Tawfiq